

ANMOL PATALAY

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Education

Maharashtra Institute of Technology, WPU - CGPA: 9.26/10

Jul 2021 – Jul 2025

Bachelor of Technology in Electronics and Communication Engineering (AI & ML Specialization)

Pune, Maharashtra

Professional Summary

Python Backend Developer building production-grade Python systems with FastAPI, SQL, and Docker. Experience designing REST APIs, authentication/authorization layers, and database-backed services, with a foundation in AI/ML from applied academic and industry projects.

Experience

Visteon Corporation

August 2025 – Present

Software Engineer

Pune, Maharashtra

- Developed reusable **Python libraries using OOP**, integrating Pandas, CAN interfaces, OpenCV, and Pytesseract to automate end-to-end cluster validation.
- Reduced CAN-based validation time by **94%**, from 3 minutes to 11 seconds, by automating signal mapping, CAN triggering, and computer-vision-based verification.
- Automated **Excel-to-Robot Framework test generation** using Python file handling and data processing, reducing manual test-authoring effort by **90%**
- Built a **FastAPI backend** that parses dynamically updated XML data and powers a live dashboard with real-time test execution and cumulative pass/fail results

Projects

Secure Job Application Tracking Backend GitHub

- Designed a **versioned REST API** with 20+ endpoints using **FastAPI**, Pydantic schemas, pagination, and OpenAPI documentation for applications, companies, and interviews.
- Modeled relational data using **MySQL, SQLAlchemy ORM, and Alembic**, with modular service-layer architecture for maintainable database-backed services.
- Implemented **JWT authentication** with OAuth2 Password Flow, bcrypt password hashing, HTTP-only refresh cookies, dependency-injected authorization guards, and user-scoped access control.
- Built secure account workflows for registration, login, password updates, and protected profiles, adding **rate limiting and automated tests** for authentication and refresh-token flows.

House Price Prediction API Live Demo

- Trained a regression model for California house price prediction and serialized it with Joblib for efficient, low-latency inference.
- Built a FastAPI backend exposing the model through REST endpoints, with Pydantic input validation and structured JSON responses for reliable real-time inference.
- Containerized the app with Docker, optimizing image builds via dependency-layer caching to reduce build time and image size.
- Deployed on Render through GitHub-integrated CI/CD, enabling automatic redeloys on every push to main.

Long-Distance couples App Live Demo

- Built a **FastAPI backend** with layered authorization dependencies, ensuring paired users can access only their shared calendar and photo resources using a couple key derived from user IDs.
- Implemented production authentication using **bcrypt**, Google ID-token verification, 30-minute access JWTs, and 1-day refresh JWTs with **SHA-256 hashing, rotation, revocation, and MongoDB TTL expiry**; added **pytest** coverage for token security and password hashing.
- Implemented server-side geolocation and media processing using **Nominatim, timezonefinder, ntplib, geopy, and Pillow**; optimized uploads to 1600px JPEGs and 480px thumbnails, with pytest validation, and deployed using **multi-stage Docker, Python 3.12, and Unicorn**.

Technical Skills

Backend/APIs: FastAPI, Flask, REST APIs, SQLAlchemy, JWT/OAuth2, Redis

Languages: Python, C++, SQL

SQL/Databases: MySQL, Complex Joins, CTEs, Window Functions, Views, Normalization, Query Optimization, Indexing, Transactions (ACID)

DevOps/Tools: Docker, Git, GitHub Actions (CI/CD), Linux CLI

AI/ML: scikit-learn, TensorFlow, PyTorch, LangChain, HuggingFace Transformers, OpenAI API, RAG Pipelines